Industrial ball mill: shell and trunnion fatigue study

MPrep Engineering - case study, October 2026. Generic overflow ball mill, 4.0 m shell ID x 6.0 m EGL, ~1.3 MW at the pinion. Parametric CAD (gmsh/OpenCASCADE), CalculiX static FEA, IIW hot-spot and Goodman fatigue assessment.

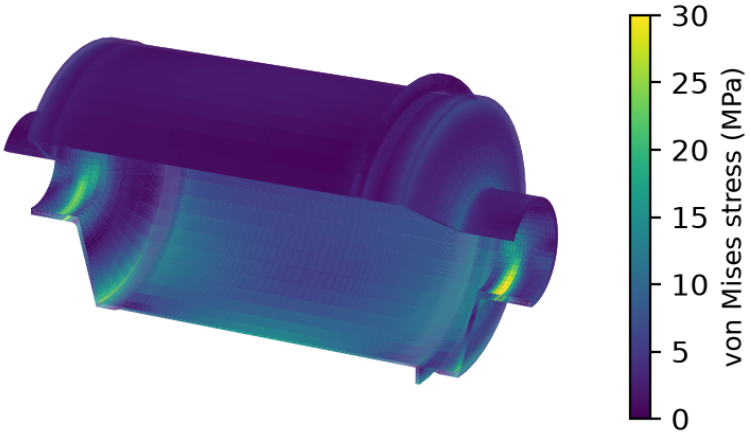


Figure 0 - Recommended design, cutaway, von Mises stress under operating load (charge, gear, self-weight and liners).

1. Summary

The baseline mill (40 mm shell, 70 mm conical heads) is statically very lightly loaded (peak von Mises 44 MPa away from the bearing pads) but **fails the 25-year fatigue requirement at the shell-to-head weld on the charge side**: the hot-spot stress range is 59 MPa per revolution, utilisation 1.64 against the IIW FAT 100 curve with gamma_M = 1.30. Each revolution is one cycle, so 25 years is 1.95 x 10^8 cycles. The driver is ovalisation of the shell under the charge: the stiff heads stop the shell end from ovalising, which forces local meridional bending at the junction. Making the heads thicker barely helps (70 -> 130 mm cuts the range by only 13 % and still leaves utilisation 1.42).

Recommended fix: 70 mm end courses, 250 mm long, with a 1:4 taper down to the 40 mm shell. This halves the hot-spot range to 28 MPa (utilisation 0.87) for +2.2 t of steel (+5 % structural mass). Thickening the whole shell to 70 mm gives the same result but costs +18.6 t. The trunnion fillet (Goodman SF 4.1) and the girth-gear flange welds (utilisation 0.78) pass in both designs. Fillet radii of 100 mm or more give no further gain.

Check	Criterion	Baseline	Recommended
Shell/head weld, hot-spot range	IIW FAT 100, f(t), gM 1.30	58.5 MPa	27.8 MPa
utilisation (25 y)	<= 1.0	**1.64 FAIL**	0.87 pass
estimated life	>= 25 years	0.4 y	> 100 y
End-course butt weld (taper end)	IIW FAT 100	-	0.44
Girth-gear flange weld	IIW FAT 90 (load carrying)	0.79	0.78
Trunnion fillet R100 (cast steel)	Goodman SF >= 2.0	4.3	4.1
Peak von Mises (excl. bearing pads)	< Sy/1.5 = 167 MPa	44 MPa	38 MPa
Max displacement	-	1.30 mm	1.26 mm
Steel mass (shell + heads + trunnions)	-	44.5 t	46.7 t

2. Design basis

Parameter	Value	Note
Shell ID / inside liners / EGL	4.00 / 3.85 / 6.0 m	75 mm average liner + lifter
Critical speed / operating speed	21.6 / 16.2 rpm	75 % Nc; Nc = 42.3/sqrt(D)
Ball filling / ball mass / slurry	35 % / 114 t / 18 t	bulk 4.65 t/m3, slurry 1.8 t/m3 in voids
Pinion power (Bond-Rowland)	1293 kW (11.4 kW/t balls)	installed motor 1500 kW
Drive torque / gear tangential / radial	764 kNm / 263 kN / 96 kN	pitch radius 2.9 m, 20 deg
Girth gear mass	25 t	distributed on flange rim
Liner areal mass	585 kg/m2	smeared into shell/head density
Cycles per year / design cycles	7.82 M / 195 M	92 % availability, 25 y
Shell, heads, trunnion (baseline)	t = 40 mm; heads 70 mm, 15 deg cone; trunnion 1400/1600 mm x 800 mm	S355 plate; cast steel trunnion/head
Charge contact arc	toe -40 deg to shoulder 135 deg from BDC	see section 3

3. Model

Geometry and mesh. The axisymmetric profile (shell, conical heads, trunnions, gear mounting flange) is defined parametrically in Python and revolved in gmsh into a conforming hexahedral mesh: C3D8I incompatible-mode bricks (a few C3D6 wedges in the heads), structured through the shell thickness, 6 mm meridional size at the junctions, 48 divisions around the circumference, 74,621 nodes. The same script exports the STEP solids.

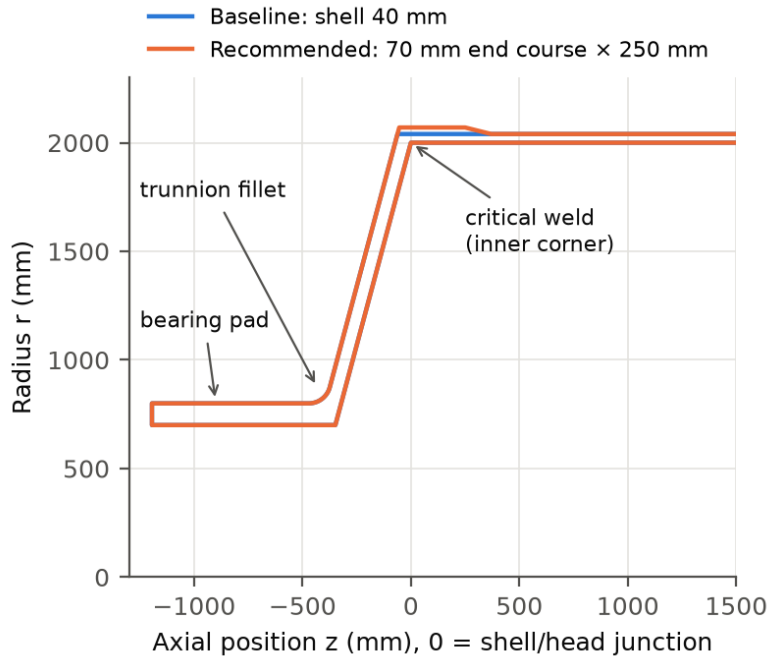


Figure 1 - Feed-end profile: baseline vs recommended end course. The discharge end is the mirror image; the gear flange sits 600 mm from the discharge junction.

Loads. (i) Self-weight with liner mass smeared into the shell and head density. (ii) Charge traction on the shell inner surface over the toe-to-shoulder arc, with a skewed profile $p = p_0 s(1-s)^b$ plus a tangential drag $q = q_0 s(1-s)^b$. The three unknowns (p_0, q_0, b) are solved so the traction exactly reproduces the charge weight, zero net horizontal force and the drive torque (result: peak 0.086 MPa near bottom dead centre, effective friction ratio 0.27). (iii) Lateral charge pressure on the head liners, K = 0.4 on the static bed. (iv) Girth-gear weight on the flange rim, plus tangential and separating mesh forces at the pinion (75 deg from BDC).

Supports. Each trunnion OD sits on a 120 deg x 400 mm hydrodynamic pad, modelled as a radial constraint in cylindrical coordinates. The feed end is the axial locating bearing and the discharge end is free to expand. One tangential constraint removes rigid-body spin; its reaction is zero because the gear torque equals the charge torque.

Fatigue method. The mill rotates through a stationary load field, so a material point at angle theta experiences the stress computed at theta: one static solution gives the full stress cycle of every axisymmetric detail. Welds: IIW type-a structural

hot-spot stress, sigma$_{hs}$ = 1.67 sigma(0.4t) - 0.67 sigma(1.0t), meridional component. Assessed on FAT 100 (full-penetration butt welds) or FAT 90 (load-carrying fillet welds), with thickness factor (25/t)^{n}, slope m = 3 up to 10^{7} cycles and m = 22 beyond, gamma$_{M}$ = 1.30. Trunnion fillet (non-welded cast steel, Sut 485 / Sy 250 MPa): Goodman with Se = 0.5 Sut x surface 0.88 x size 0.60 x reliability 0.81 x casting quality 0.80 = 83 MPa.

4. Results

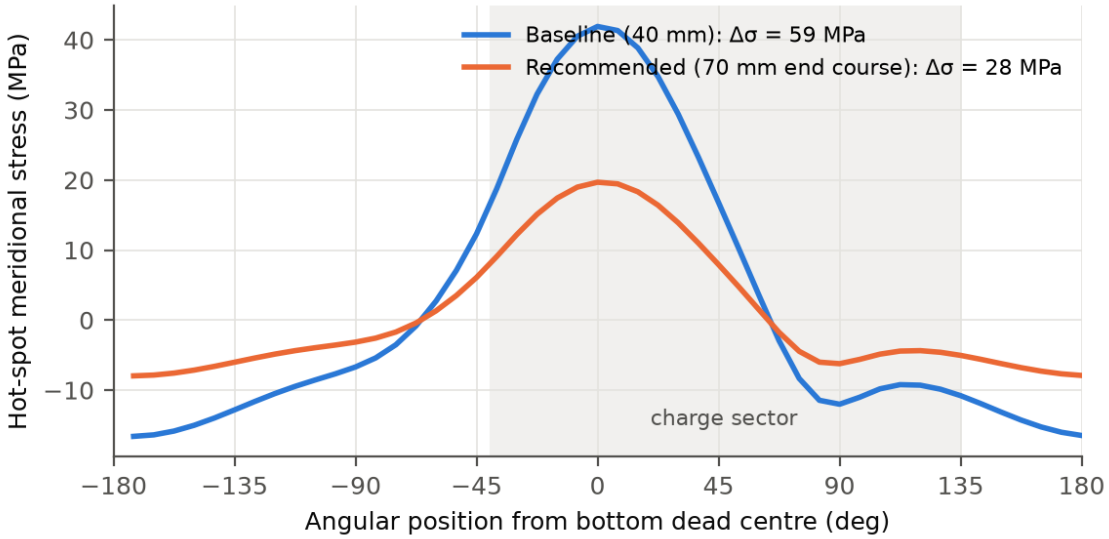


Figure 2 - Hot-spot stress over one revolution at the discharge-end shell/head weld (inner surface). The peak occurs as the joint passes under the charge.

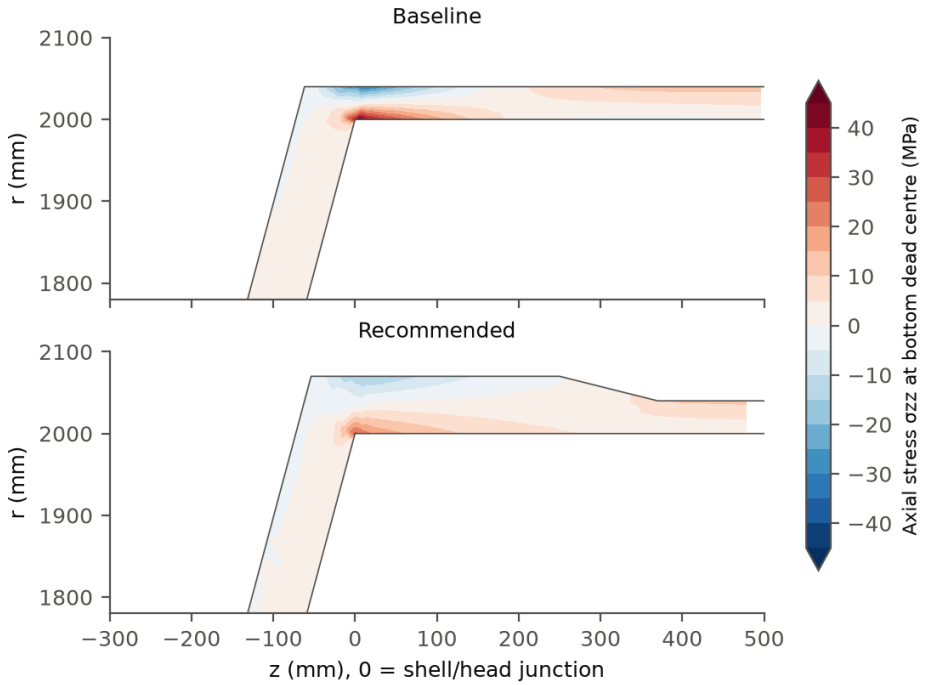


Figure 3 - Axial stress in the meridional section at bottom dead centre, feed-end junction. Tension concentrates at the charge-side inner corner; the end course spreads and reduces it.

5. Design study

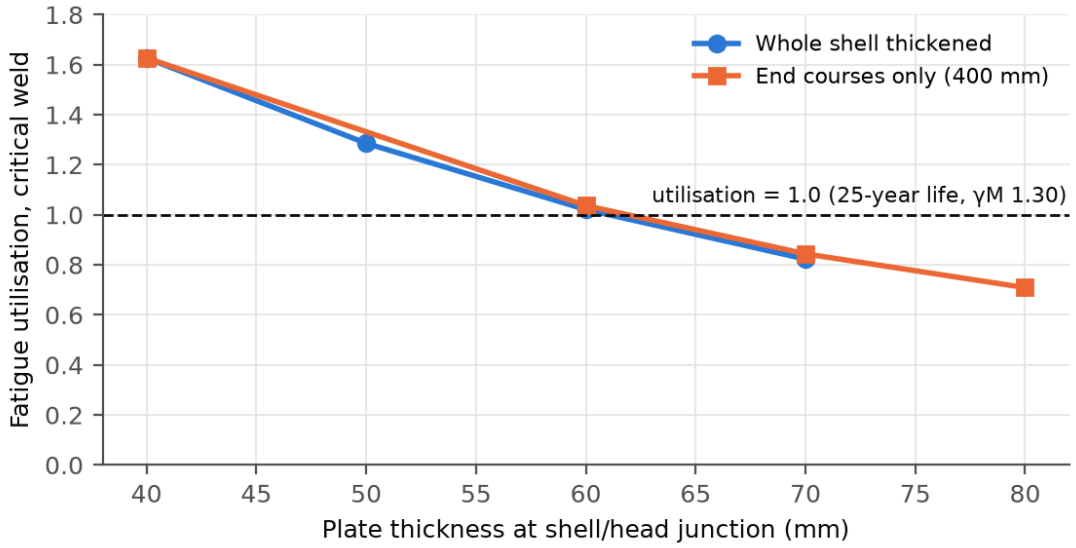


Figure 4 - Fatigue utilisation of the critical weld vs plate thickness at the junction. Thickening only the end courses performs the same as thickening the whole shell, at about one eighth of the added steel.

Variant	Delta sigma_hs (MPa)	Utilisation	Added steel
Baseline 40 mm	58.2	1.63	-
Heads 90 / 110 / 130 mm	57.3 / 54.0 / 50.9	1.60 / 1.51 / 1.42	+3.8 / +7.5 / +11.3 t
Whole shell 50 / 60 / 70 mm	44.0 / 33.7 / 26.3	1.29 / 1.02 / 0.82	+6.2 / +12.4 / +18.6 t
End courses 60 / 70 / 80 mm x 400 mm	34.2 / 27.0 / 22.1	1.04 / 0.84 / 0.71	+2.0 / +3.2 / +4.4 t
End courses 70 mm x 250 mm (rec.)	27.5	**0.86**	+2.2 t

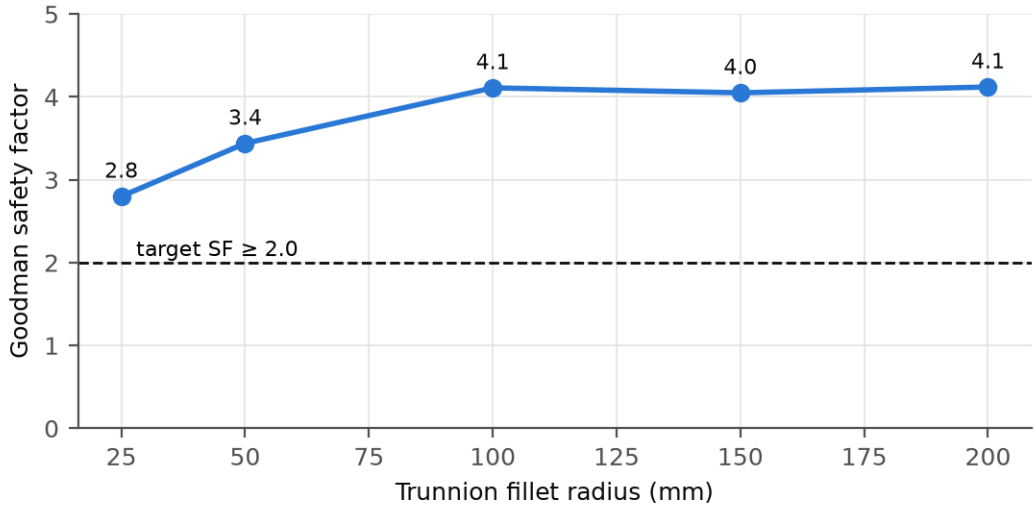
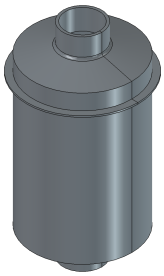


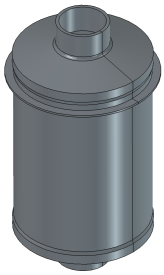
Figure 5 - Trunnion fillet Goodman safety factor vs fillet radius (recommended shell). The fillet is not life-limiting; 100 mm is enough.

6. Model and results in the native tools

The figures below are taken directly from the tools used for the study: FreeCAD 1.1 for the CAD geometry (STEP exports of both designs) and PrePoMax 3.0 (CalculiX post-processor) for the stress results. All PrePoMax views show S33, the global axial stress, which is the meridional stress on the shell surface used for the IIW hot-spot read-out. Both designs use the same section plane (through the mill axis, x = 0), the same camera, undeformed geometry and the same fixed legend of -50 to +50 MPa, so colours compare directly. The discharge end is on the left in every PrePoMax view; the bottom of the section is the charge side (theta = 0), where the shell/head weld sees its peak tension.

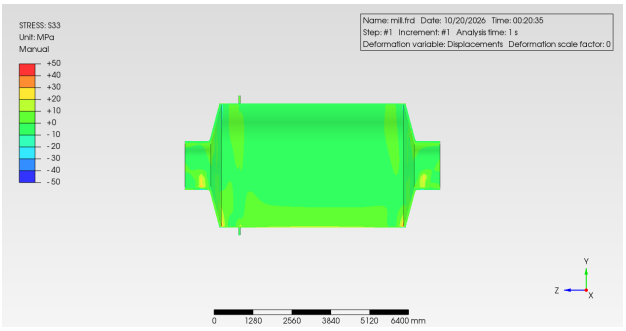


Baseline: 40 mm shell up to the head junction.

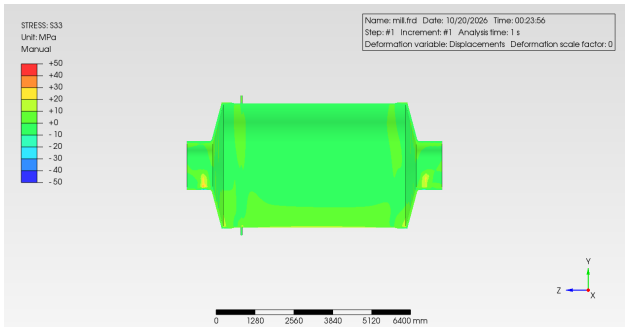


Recommended: 70 mm end courses with 1:4 taper (visible as rings at each end).

Figure 6 - CAD models of both designs, FreeCAD 1.1, isometric view.

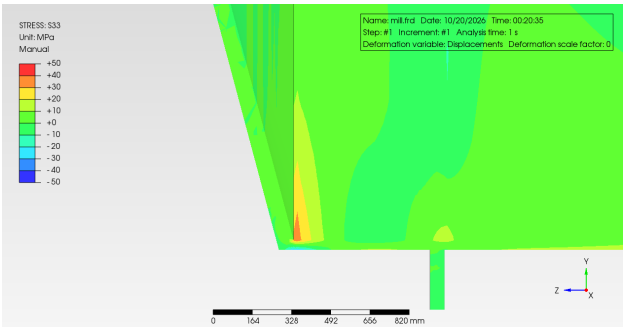


Baseline

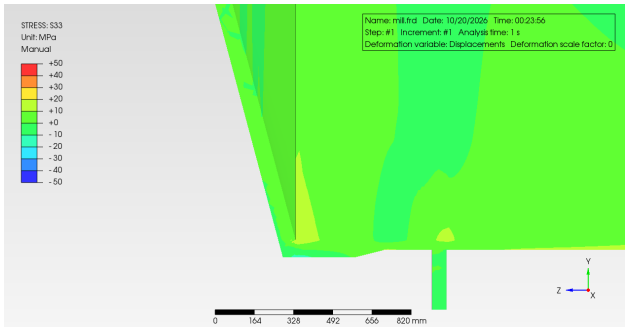


Recommended

Figure 7 - S33 (MPa) on the axial section through the whole mill, PrePoMax. The shell is lightly loaded everywhere except near the shell/head junctions.



Baseline: hot-spot range 58.5 MPa (S_hs 41.9 / -16.6 MPa).



Recommended: hot-spot range 27.6 MPa (S_hs 19.7 / -8.0 MPa).

Figure 8 - Close-up of the discharge-end inner shell/head weld on the charge side, PrePoMax. The orange-red tension spot at the baseline weld toe drops to light yellow with the thicker end course.

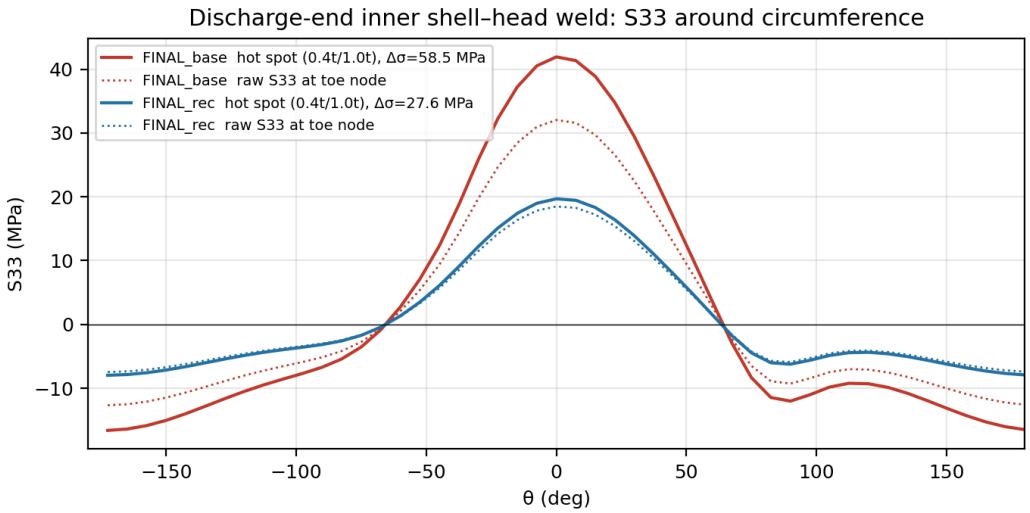


Figure 9 - S33 around the circumference at the discharge-end inner weld (independent re-read of the CalculiX results): IIW hot-spot stress (0.4t / 1.0t extrapolation) and the raw toe-node value. The script reproduces the post-processing ranges exactly.

7. Verification

Equilibrium. Total vertical bearing reaction 2.28 MN = charge on shell 1.29 MN + gear 0.25 MN + structure and liners 1.07 MN + radial component of head pressure 0.03 MN - pinion uplift 0.35 MN. Net horizontal charge force 0 N and net torque 0.00 kNm by construction. Horizontal reaction 6.2 kN, matching the net horizontal pinion force (86 kN tangential component - 93 kN separating component).

Mesh convergence. The critical hot-spot range converges to within 0.5 % below 8 mm meridional size, and circumferential refinement from 48 to 60 divisions changes it by under 0.1 %. A final refinement to 4 mm and 72 divisions (142,445 / 126,536 nodes) moves the weld range by -1.2 % (baseline) and +0.8 % (recommended): utilisation 1.62 / 0.88, so neither verdict changes. The trunnion fillet is the most mesh-sensitive detail: on the recommended design its Goodman SF drops from 4.1 to 3.5 at the finest mesh, still well above 2.0.

Mesh (junction size / circumf. div.)	12.5 mm / 48	12.5 mm / 60	8 mm / 36	8 mm / 48	6 mm / 36	6 mm / 48	4 mm / 72
Weld Delta sigma_hs (MPa)	60.3	60.3	58.1	58.4	58.2	58.5	57.8
Fillet sigma_a (MPa)	16.8	16.8	17.8	17.8	18.5	18.6	16.9

Recommended design	6 mm / 48	4 mm / 72
Weld Delta sigma_hs (MPa)	27.8	28.0
utilisation	0.870	0.876
Fillet Goodman SF	4.1	3.5

Solver cross-check. The baseline solved on CalculiX 2.17 (desktop) and 2.21 (cloud) agrees within 1 %. Re-running both final models on CalculiX 2.22 (PrePoMax build, Windows) reproduces every hot-spot value to 0.01 MPa.

8. Limitations and next steps

The charge pressure profile is a calibrated engineering distribution, not a DEM result. A DEM charge-motion run (e.g. 3D, 75 % Nc) would refine the toe and shoulder loads. Welds are not modelled geometrically; the shell/head inner corner is sharp, and the IIW read-out points (16 / 40 mm) sit outside the corner singularity zone. Manholes, bolted head flanges, liner bolt holes, thermal gradients, start-up and inching torque (typically 1.5-2.0x), and misalignment of the gear mesh are not included. They are the obvious next layer for a project-specific assessment, together with a submodel of the manhole cut-out.

References

[1] Hobbacher, A.F. (2016). Recommendations for Fatigue Design of Welded Joints and Components, IIW doc. XIII-2460-13, Springer. [2] Rowland, C.A. and Kjos, D.M. (1980). Rod and ball mills, in Mineral Processing Plant Design, SME. [3] Budynas, R.G. and Nisbett, J.K. Shigley's Mechanical Engineering Design, 10th ed., McGraw-Hill (Marin factors, Goodman). [4] Dhondt, G. CalculiX CrunchiX User's Manual, v2.17-2.22.